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Student ID: 24B031086

Group: Tuesday (12:00-14:00)

Part 1: Setting Up a Simple DHCP Network

1. Router Configuration:

Router0

Physical Config CLI Attributes

IOS Command Line Interface

```
         --- System Configuration Dialog ---

Would you like to enter the initial configuration dialog? [yes/no]: no


Press RETURN to get started!



Router>en
Router#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#int fa0/0
Router(config-if)#no sh

Router(config-if)#
%LINK-5-CHANGED: Interface FastEthernet0/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0, changed
state to up

Router(config-if)#ip address 192.168.1.1 255.255.255.0
Router(config-if)#
```

2.Configuring DHCP Pool on Router:

Router0

Physical Config CLI Attributes

IOS Command Line Interface

```
Press RETURN to get started!



Router>en
Router#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#int fa0/0
Router(config-if)#no sh

Router(config-if)#
%LINK-5-CHANGED: Interface FastEthernet0/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0, changed
state to up

Router(config-if)#ip address 192.168.1.1 255.255.255.0
Router(config-if)#ip dhcp pool DHCP
Router(dhcp-config)#network 192.168.1.0 255.255.255.0
Router(dhcp-config)#defa
Router(dhcp-config)#default-router 192.168.1.1
Router(dhcp-config)#dns
Router(dhcp-config)#dns-server 8.8.8.8
Router(dhcp-config)#exit
Router(config)#
```

3.Excluding IP Addresses:

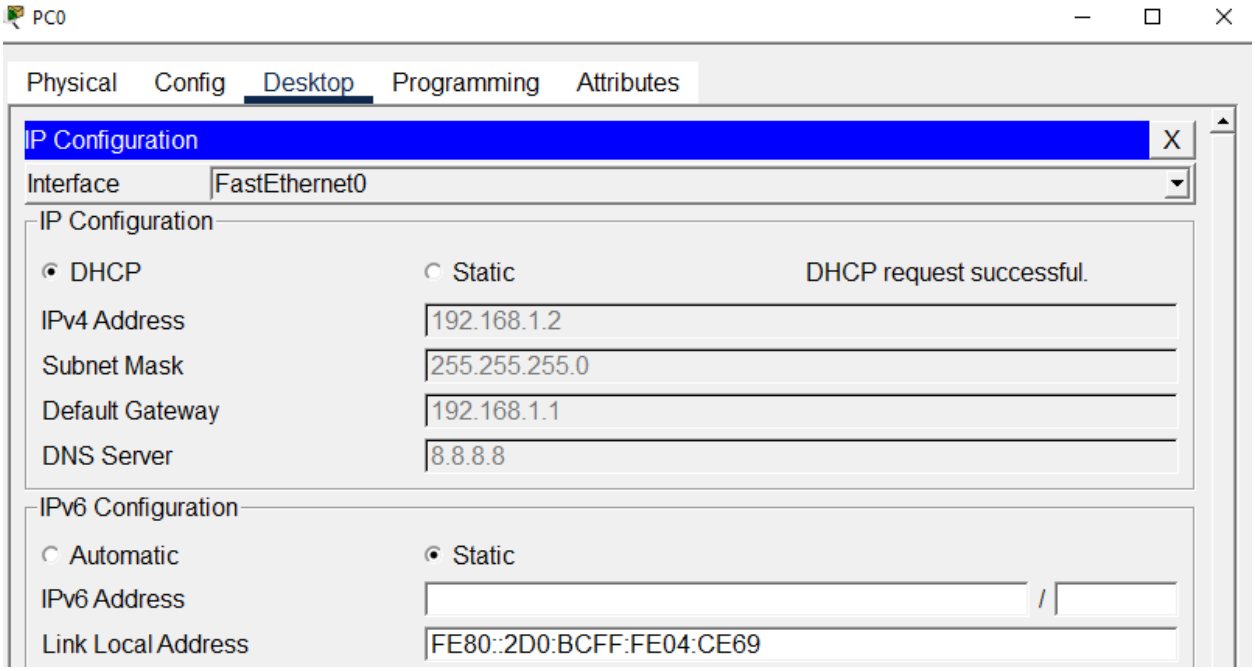
Router0

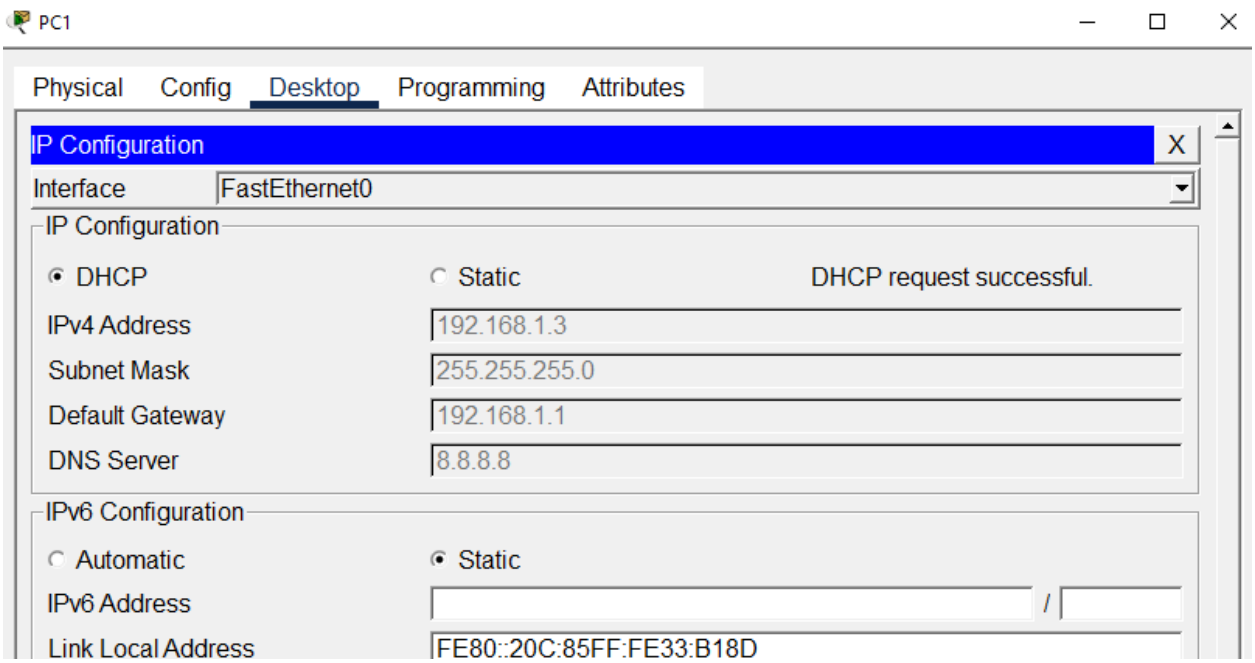
Physical Config CLI Attributes

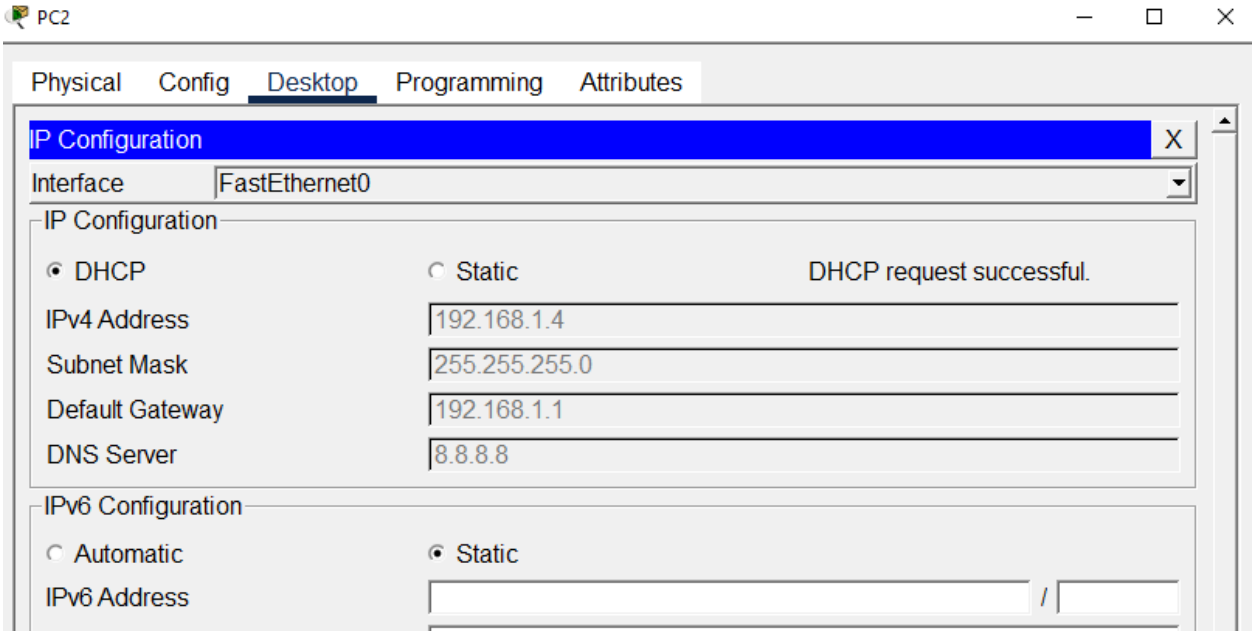
IOS Command Line Interface

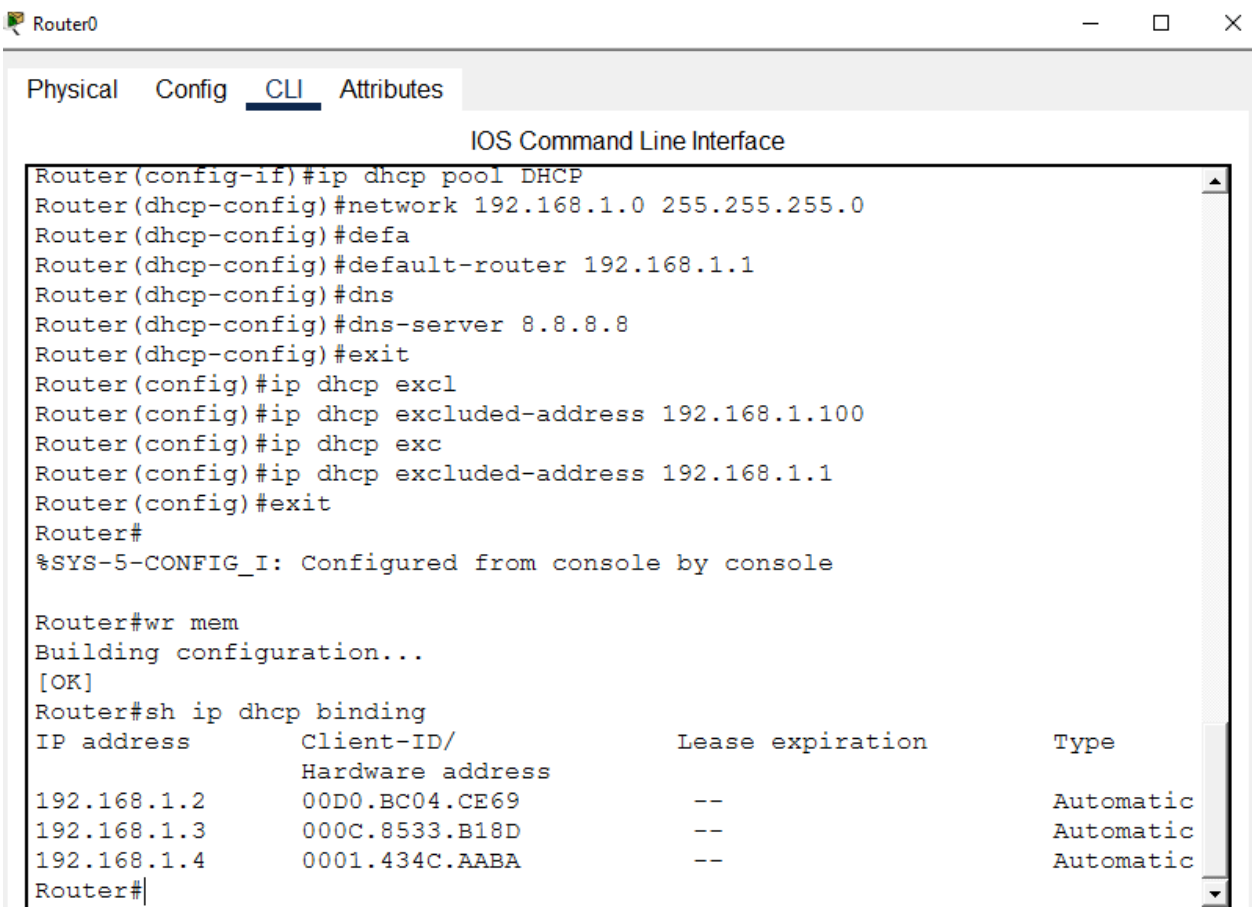
```
%LINK-5-CHANGED: Interface FastEthernet0/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0, changed
state to up

Router(config-if)#ip address 192.168.1.1 255.255.255.0
Router(config-if)#ip dhcp pool DHCP
Router(dhcp-config)#network 192.168.1.0 255.255.255.0
Router(dhcp-config)#defa
Router(dhcp-config)#default-router 192.168.1.1
Router(dhcp-config)#dns
Router(dhcp-config)#dns-server 8.8.8.8
Router(dhcp-config)#exit
Router(config)#ip dhcp excl
Router(config)#ip dhcp excluded-address 192.168.1.100
Router(config)#ip dhcp exc
Router(config)#ip dhcp excluded-address 192.168.1.1
Router(config)#exit
Router#
%SYS-5-CONFIG_I: Configured from console by console

Router#wr mem
Building configuration...
[OK]
Router#
```

4. PC Configuration:









5. Test Connectivity:

PC0

Physical Config Desktop Programming Attributes

Command Prompt

```
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 192.168.1.3

Pinging 192.168.1.3 with 32 bytes of data:

Reply from 192.168.1.3: bytes=32 time<1ms TTL=128
Reply from 192.168.1.3: bytes=32 time<1ms TTL=128
Reply from 192.168.1.3: bytes=32 time<1ms TTL=128
Reply from 192.168.1.3: bytes=32 time<1ms TTL=128

Ping statistics for 192.168.1.3:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>ping 192.168.1.1

Pinging 192.168.1.1 with 32 bytes of data:

Reply from 192.168.1.1: bytes=32 time<1ms TTL=255
Reply from 192.168.1.1: bytes=32 time=1ms TTL=255
Reply from 192.168.1.1: bytes=32 time<1ms TTL=255
Reply from 192.168.1.1: bytes=32 time=1ms TTL=255

Ping statistics for 192.168.1.1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
```

PC0

Physical Config Desktop Programming Attributes

Command Prompt

```
Pinging 192.168.1.3 with 32 bytes of data:

Reply from 192.168.1.3: bytes=32 time<1ms TTL=128
Reply from 192.168.1.3: bytes=32 time<1ms TTL=128
Reply from 192.168.1.3: bytes=32 time<1ms TTL=128
Reply from 192.168.1.3: bytes=32 time<1ms TTL=128

Ping statistics for 192.168.1.3:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>ping 192.168.1.1

Pinging 192.168.1.1 with 32 bytes of data:

Reply from 192.168.1.1: bytes=32 time<1ms TTL=255
Reply from 192.168.1.1: bytes=32 time=1ms TTL=255
Reply from 192.168.1.1: bytes=32 time<1ms TTL=255
Reply from 192.168.1.1: bytes=32 time=1ms TTL=255

Ping statistics for 192.168.1.1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 1ms, Average = 0ms

C:\>
```

Part 2: VLAN Configuration with DHCP Relay

1. VLAN Configuration on the Switch, create VLANs and Assign Names:

Switch1

Physical Config CLI Attributes

IOS Command Line Interface

```
%LINK-5-CHANGED: Interface FastEthernet0/5, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/5, changed
state to up

%LINK-5-CHANGED: Interface FastEthernet0/6, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/6, changed
state to up


Switch>en
Switch#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
Switch(config)#vlan 2
Switch(config-vlan)#name VLAN2
Switch(config-vlan)#exit
Switch(config)#vlan 3
Switch(config-vlan)#name VLAN3
Switch(config-vlan)#exit
Switch(config)#vlan 4
Switch(config-vlan)#name DHCP
Switch(config-vlan)#exit
Switch(config)#
```

2. Assign VLANs to Switch Ports:

Switch1

Physical Config CLI Attributes

IOS Command Line Interface

```
Switch(config)#vlan 3
Switch(config-vlan)#name VLAN3
Switch(config-vlan)#exit
Switch(config)#vlan 4
Switch(config-vlan)#name DHCP
Switch(config-vlan)#exit
Switch(config)#int range fa0/2-3
Switch(config-if-range)#swit
Switch(config-if-range)#switchport mode access
Switch(config-if-range)#swit
Switch(config-if-range)#switchport access vlan 2
Switch(config-if-range)#exit
Switch(config)#int range fa0/4-5
Switch(config-if-range)#swit
Switch(config-if-range)#switchport mode access
Switch(config-if-range)#swi
Switch(config-if-range)#switchport access vlan 3
Switch(config-if-range)#exit
Switch(config)#int fa0/6
Switch(config-if)#swit
Switch(config-if)#switchport mode access
Switch(config-if)#swi
Switch(config-if)#switchport access vlan 4
Switch(config-if)#exit
Switch(config)#
```

3.Configure the Trunk Port:

Switch1

Physical Config CLI Attributes

IOS Command Line Interface

```
Switch(config)#int range fa0/4-5
Switch(config-if-range)#swit
Switch(config-if-range)#switchport mode access
Switch(config-if-range)#swi
Switch(config-if-range)#switchport access vlan 3
Switch(config-if-range)#exit
Switch(config)#int fa0/6
Switch(config-if)#swit
Switch(config-if)#switchport mode access
Switch(config-if)#swi
Switch(config-if)#switchport access vlan 4
Switch(config-if)#exit
Switch(config)#int fa0/1
Switch(config-if)#swit
Switch(config-if)#switchport mode trunk
Switch(config-if)#swit
Switch(config-if)#switchport trunk allowed vlan 2,3,4
Switch(config-if)#exit
Switch(config)#end
Switch#
%SYS-5-CONFIG_I: Configured from console by console
wr mem
Building configuration...
[OK]
Switch#
```

4. Configuring Sub-Interfaces on the Router for Inter- VLAN Routing:

Router1

Physical Config CLI Attributes

IOS Command Line Interface

```
Router>en
Router#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#int g0/0.2
Router(config-subif)#enca
Router(config-subif)#encapsulation dot
Router(config-subif)#encapsulation dot1Q 2
Router(config-subif)#ip address 192.168.2.1 255.255.255.0
Router(config-subif)#no sh
Router(config-subif)#exit
Router(config)#int g0/0.3
Router(config-subif)#encap
Router(config-subif)#encapsulation dot
Router(config-subif)#encapsulation dot1Q 3
Router(config-subif)#ip address 192.168.3.1 255.255.255.0
Router(config-subif)#no sh
Router(config-subif)#exit
Router(config)#int g0/0.4
Router(config-subif)#enca
Router(config-subif)#encapsulation dot
Router(config-subif)#encapsulation dot1Q 4
Router(config-subif)#ip address 192.168.4.1 255.255.255.0
Router(config-subif)#no sh
Router(config-subif)#exit
Router(config)#
```

Router1

Physical Config CLI Attributes

IOS Command Line Interface

```
Router(config)#int g0/0
Router(config-if)#no sh

Router(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0,
changed state to up

%LINK-3-UPDOWN: Interface GigabitEthernet0/0.2, changed state to down

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0.2,
changed state to up

%LINK-3-UPDOWN: Interface GigabitEthernet0/0.3, changed state to down

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0.3,
changed state to up

%LINK-3-UPDOWN: Interface GigabitEthernet0/0.4, changed state to down

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0.4,
changed state to up
end
Router#
```

Copy Paste

Router1

Physical Config CLI Attributes

IOS Command Line Interface

```
%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0,
changed state to up

%LINK-3-UPDOWN: Interface GigabitEthernet0/0.2, changed state to down

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0.2,
changed state to up

%LINK-3-UPDOWN: Interface GigabitEthernet0/0.3, changed state to down

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0.3,
changed state to up

%LINK-3-UPDOWN: Interface GigabitEthernet0/0.4, changed state to down

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0.4,
changed state to up
end
Router#
%SYS-5-CONFIG_I: Configured from console by console
wr mem
Building configuration...
[OK]
Router#
```

5.Configure DHCP Server:

Server0

Physical | Config | Services | Desktop | Programming | Attributes

IP Configuration

IP Configuration

DHCP	Static
IPv4 Address	192.168.4.2
Subnet Mask	255.255.255.0
Default Gateway	192.168.4.1
DNS Server	0.0.0.0

IPv6 Configuration

Automatic	Static
IPv6 Address	/
Link Local Address	FE80::2D0:58FF:FE6B:D647

Server0

Physical | Config | Services | Desktop | Programming | Attributes

SERVICES

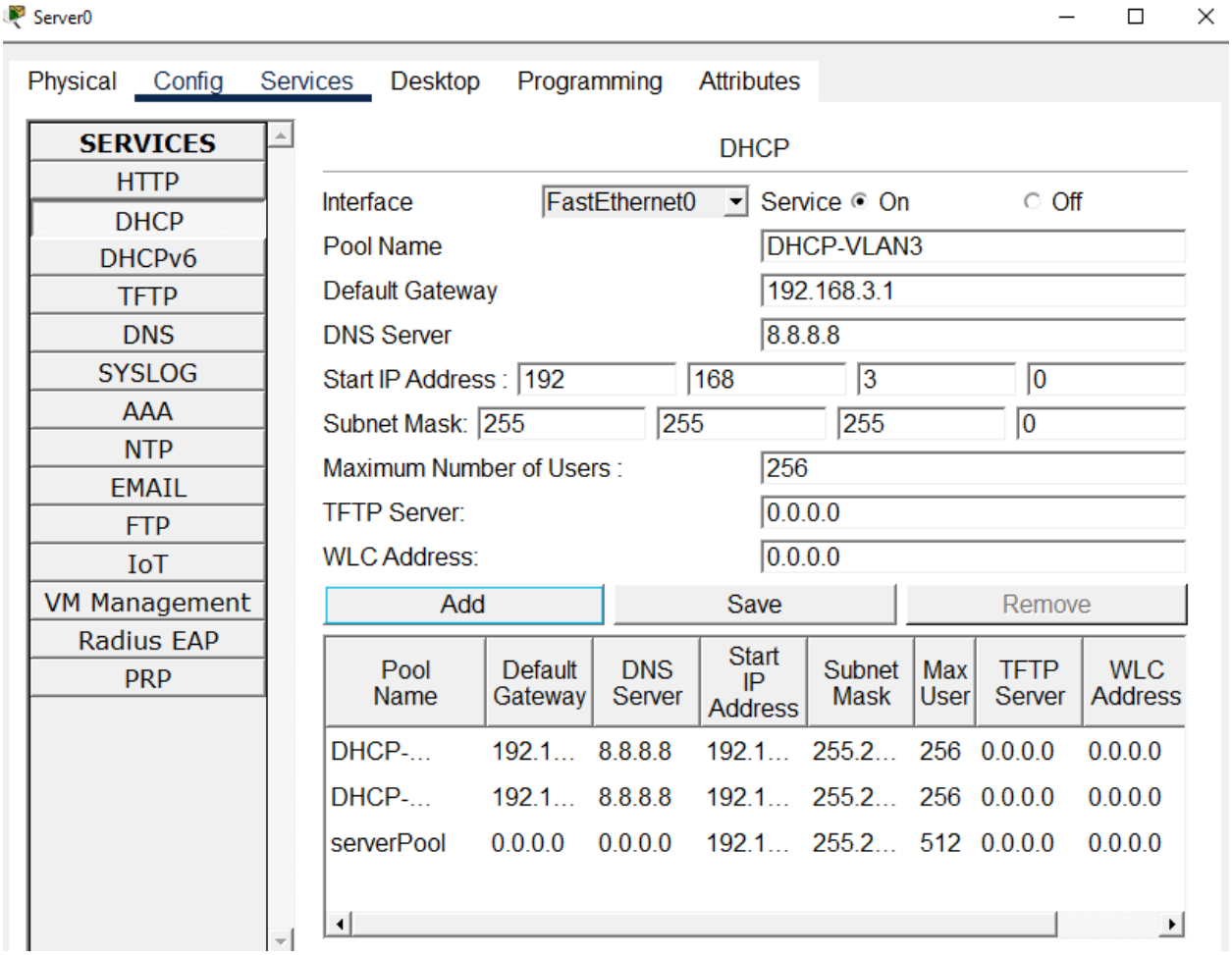
- HTTP
- DHCP
- DHCPv6
- TFTP
- DNS
- SYSLOG
- AAA
- NTP
- EMAIL
- FTP
- IoT
- VM Management
- Radius EAP
- PRP

DHCP

Interface	FastEthernet0
Service	On / Off
Pool Name	DHCP-VLAN2
Default Gateway	192.168.2.1
DNS Server	8.8.8.8
Start IP Address :	192 168 2 0
Subnet Mask:	255 255 255 0
Maximum Number of Users :	256
TFTP Server:	0.0.0.0
WLC Address:	0.0.0.0

Add | Save | Remove

Pool Name	Default Gateway	DNS Server	Start IP Address	Subnet Mask	Max User	TFTP Server	WLC Address
DHCP-...	192.1...	8.8.8.8	192.1...	255.2...	256	0.0.0.0	0.0.0.0
serverPool	0.0.0.0	0.0.0.0	192.1...	255.2...	512	0.0.0.0	0.0.0.0



6.DHCP Relay Configuration:

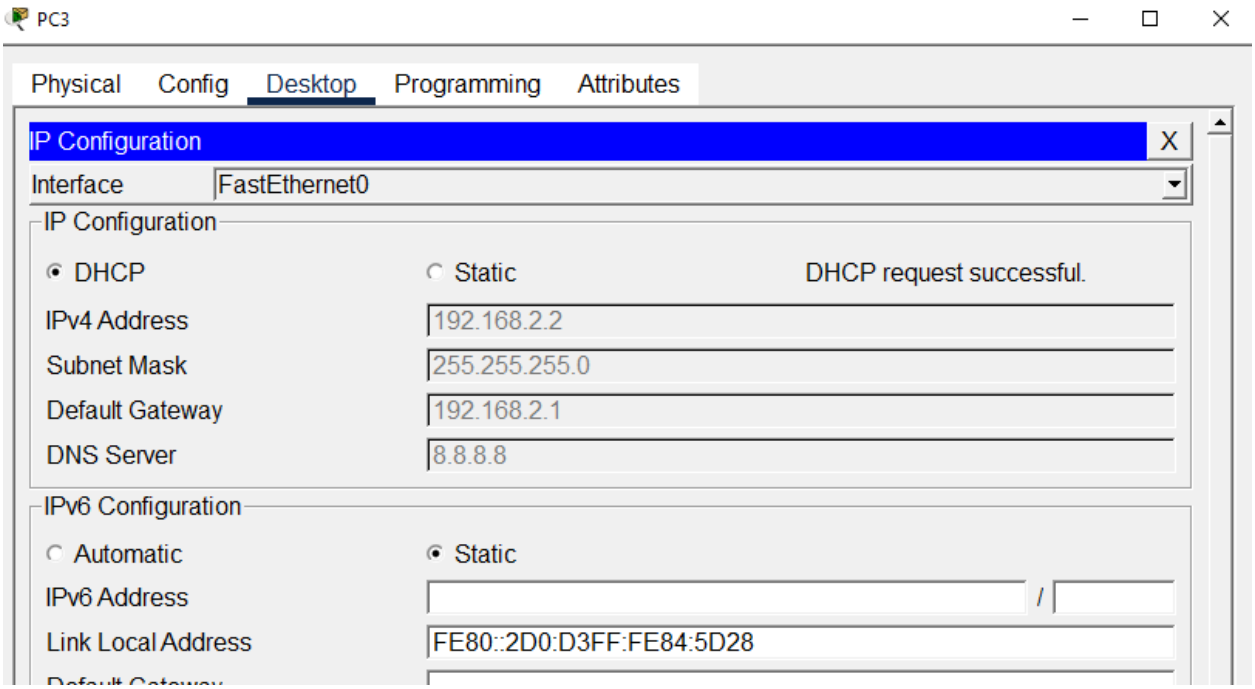
Router1

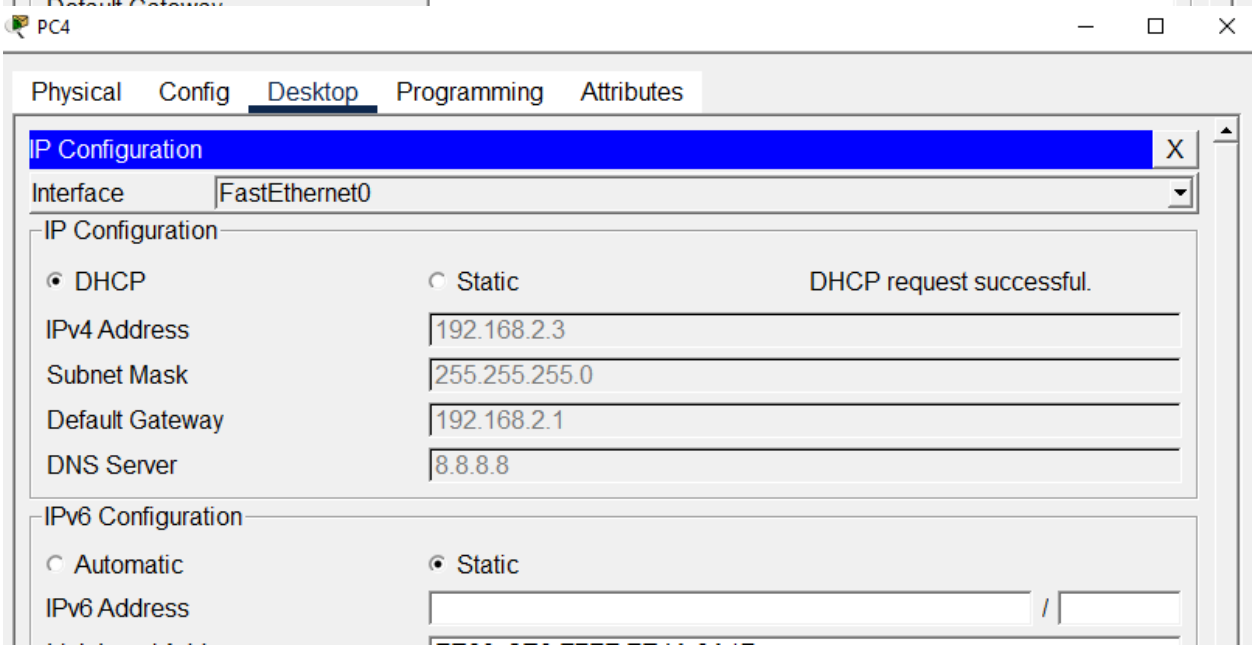
Physical Config CLI Attributes

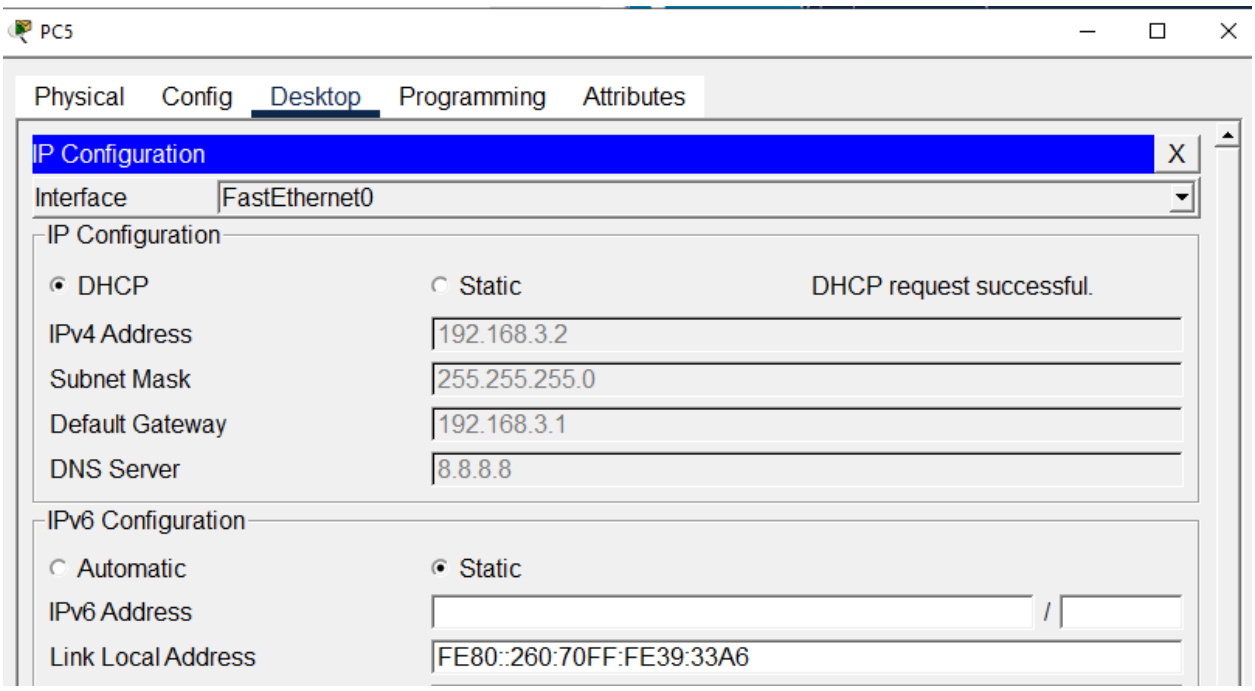
IOS Command Line Interface

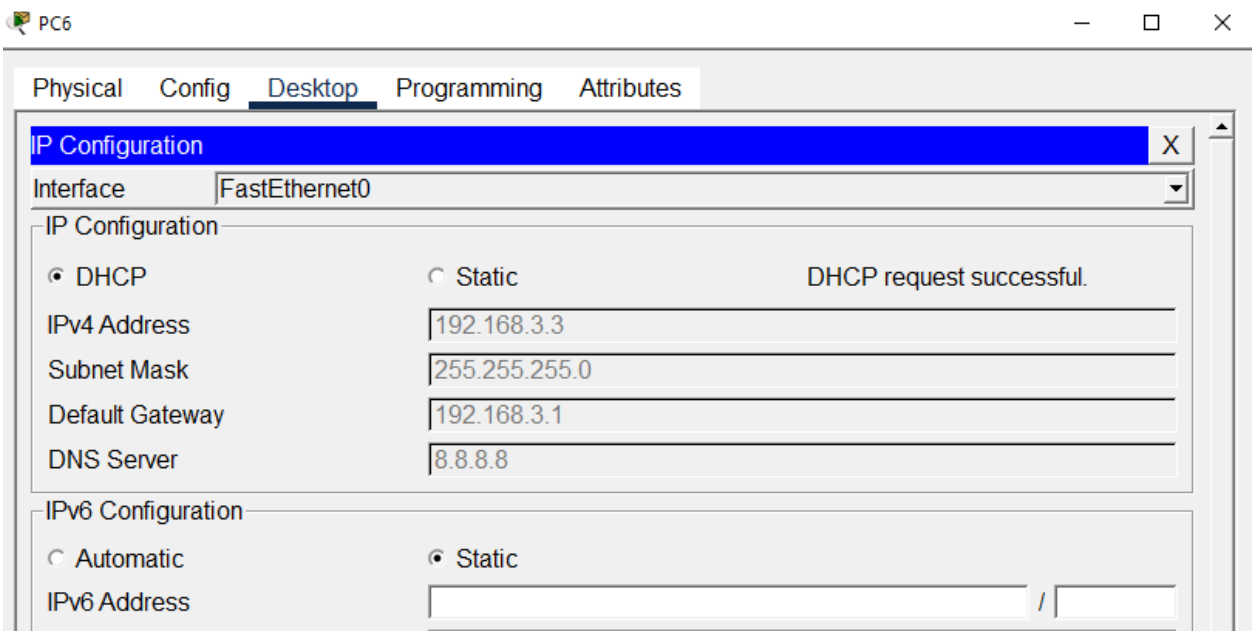
```
%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0.3,
changed state to up

%LINK-3-UPDOWN: Interface GigabitEthernet0/0.4, changed state to down

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0.4,
changed state to up
end
Router#
%SYS-5-CONFIG_I: Configured from console by console
wr mem
Building configuration...
[OK]
Router#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#int g0/0.2
Router(config-subif)#ip help
Router(config-subif)#ip helper-address 192.168.4.2
Router(config-subif)#exit
Router(config)#int g0/0.3
Router(config-subif)#ip hel
Router(config-subif)#ip help
Router(config-subif)#ip helper-address 192.168.4.2
Router(config-subif)#exit
Router(config)#
```

7. Configuration IP Addresses on PCs:


PC3
Physical
Config
Desktop
Programming
Attributes
IP Configuration
X
Interface
FastEthernet0
IP Configuration
DHCP
Static
DHCP request successful.
IPv4 Address
192.168.2.2
Subnet Mask
255.255.255.0
Default Gateway
192.168.2.1
DNS Server
8.8.8.8
IPv6 Configuration
Automatic
Static
IPv6 Address
/
Link Local Address
FE80::2D0:D3FF:FE84:5D28


PC4
Physical
Config
Desktop
Programming
Attributes
IP Configuration
X
Interface
FastEthernet0
IP Configuration
DHCP
Static
DHCP request successful.
IPv4 Address
192.168.2.3
Subnet Mask
255.255.255.0
Default Gateway
192.168.2.1
DNS Server
8.8.8.8
IPv6 Configuration
Automatic
Static
IPv6 Address
/





8. Test Connectivity:

PC3

Physical Config Desktop Programming Attributes

Command Prompt

```
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 192.168.2.3

Pinging 192.168.2.3 with 32 bytes of data:

Reply from 192.168.2.3: bytes=32 time<1ms TTL=128
Reply from 192.168.2.3: bytes=32 time<1ms TTL=128
Reply from 192.168.2.3: bytes=32 time<1ms TTL=128
Reply from 192.168.2.3: bytes=32 time<1ms TTL=128

Ping statistics for 192.168.2.3:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>
```

PC4

Physical Config Desktop Programming Attributes

Command Prompt

```
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 192.168.3.2

Pinging 192.168.3.2 with 32 bytes of data:

Request timed out.
Reply from 192.168.3.2: bytes=32 time=1ms TTL=127
Reply from 192.168.3.2: bytes=32 time<1ms TTL=127
Reply from 192.168.3.2: bytes=32 time<1ms TTL=127

Ping statistics for 192.168.3.2:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 1ms, Average = 0ms

C:\>
```

PC5

Physical Config Desktop Programming Attributes

Command Prompt

```
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 192.168.3.3

Pinging 192.168.3.3 with 32 bytes of data:

Reply from 192.168.3.3: bytes=32 time<1ms TTL=128
Reply from 192.168.3.3: bytes=32 time=1ms TTL=128
Reply from 192.168.3.3: bytes=32 time<1ms TTL=128
Reply from 192.168.3.3: bytes=32 time<1ms TTL=128

Ping statistics for 192.168.3.3:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 1ms, Average = 0ms

C:\>
```

PC6

Physical Config Desktop Programming Attributes

Command Prompt

```
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 192.168.2.2

Pinging 192.168.2.2 with 32 bytes of data:

Request timed out.
Reply from 192.168.2.2: bytes=32 time=2ms TTL=127
Reply from 192.168.2.2: bytes=32 time=1ms TTL=127
Reply from 192.168.2.2: bytes=32 time=1ms TTL=127

Ping statistics for 192.168.2.2:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
Approximate round trip times in milli-seconds:
    Minimum = 1ms, Maximum = 2ms, Average = 1ms

C:\>
```

Part 3: Additional Task - Add new VLAN

Switch1

Physical Config CLI Attributes

IOS Command Line Interface

```
%LINK-5-CHANGED: Interface FastEthernet0/7, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/7, changed
state to up

%LINK-5-CHANGED: Interface FastEthernet0/8, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/8, changed
state to up


Switch>en
Switch#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
Switch(config)#vlan 5
Switch(config-vlan)#name VLAN5
Switch(config-vlan)#exit
Switch(config)#int range fa0/7-8
Switch(config-if-range)#swit
Switch(config-if-range)#switchport mode access
Switch(config-if-range)#swi
Switch(config-if-range)#switchport access vlan 5
Switch(config-if-range)#exit
Switch(config)#
```

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Router1

Physical Config CLI Attributes

IOS Command Line Interface

```
Router(config-subif)#exit
Router(config)#int g0/0.3
Router(config-subif)#ip hel
Router(config-subif)#ip help
Router(config-subif)#ip helper-address 192.168.4.2
Router(config-subif)#exit
Router(config)#end
Router#
%SYS-5-CONFIG_I: Configured from console by console

Router#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#int g0/0.5
Router(config-subif)#
%LINK-3-UPDOWN: Interface GigabitEthernet0/0.5, changed state to down

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0.5,
changed state to up

Router(config-subif)#enca
Router(config-subif)#encapsulation dot
Router(config-subif)#encapsulation dot1Q 5
Router(config-subif)#ip address 192.168.5.1 255.255.255.0
Router(config-subif)#exit
Router(config)#
```

Copy Paste

Server0

Physical | Config | Services | Desktop | Programming | Attributes

SERVICES

- HTTP
- DHCP
- DHCPv6
- TFTP
- DNS
- SYSLOG
- AAA
- NTP
- EMAIL
- FTP
- IoT
- VM Management
- Radius EAP
- PRP

DHCP

Interface: FastEthernet0 — Service: On / Off

Pool Name: DHCP-VLAN5

Default Gateway: 192.168.5.1

DNS Server: 8.8.8.8

Start IP Address : 192 . 168 . 5 . 0

Subnet Mask: 255 . 255 . 255 . 0

Maximum Number of Users : 256

TFTP Server: 0.0.0.0

WLC Address: 0.0.0.0

Add | Save | Remove

Pool Name	Default Gateway	DNS Server	Start IP Address	Subnet Mask	Max User	TFTP Server	WLC Address
DHCP-...	192.1...	8.8.8.8	192.1...	255.2...	256	0.0.0.0	0.0.0.0
DHCP-...	192.1...	8.8.8.8	192.1...	255.2...	256	0.0.0.0	0.0.0.0
DHCP-...	192.1...	8.8.8.8	192.1...	255.2...	256	0.0.0.0	0.0.0.0
serverPool	0.0.0.0	0.0.0.0	192.1...	255.2...	512	0.0.0.0	0.0.0.0

Router1

Physical | Config | CLI | Attributes

IOS Command Line Interface

```
Router(config-subif)#ip helper-address 192.168.4.2
Router(config-subif)#exit
Router(config)#end
Router#
%SYS-5-CONFIG_I: Configured from console by console

Router#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#int g0/0.5
Router(config-subif)#
%LINK-3-UPDOWN: Interface GigabitEthernet0/0.5, changed state to down

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0.5,
changed state to up

Router(config-subif)#enca
Router(config-subif)#encapsulation dot
Router(config-subif)#encapsulation dot1Q 5
Router(config-subif)#ip address 192.168.5.1 255.255.255.0
Router(config-subif)#exit
Router(config)#int g0/0.5
Router(config-subif)#ip helper
Router(config-subif)#ip helper-address 192.168.4.2
Router(config-subif)#exit
Router(config)#
```

Copy | Paste

PC7

Physical | Config | Desktop | Programming | Attributes

IP Configuration

Interface: FastEthernet0

IP Configuration

- DHCP
- Static

DHCP request successful.

IPv4 Address	192.168.5.2
Subnet Mask	255.255.255.0
Default Gateway	192.168.5.1
DNS Server	8.8.8.8

IPv6 Configuration

- Automatic
- Static

IPv6 Address	/
Link Local Address	FE80::201:64FF:FE72:885A

PC8

Physical | Config | Desktop | Programming | Attributes

IP Configuration

Interface: FastEthernet0

IP Configuration

- DHCP
- Static

DHCP request successful.

IPv4 Address	192.168.5.3
Subnet Mask	255.255.255.0
Default Gateway	192.168.5.1
DNS Server	8.8.8.8

IPv6 Configuration

- Automatic
- Static

IPv6 Address	/
Link Local Address	FE80::2E0:A3FF:FE17:B374

PC7

Physical Config Desktop Programming Attributes

Command Prompt

```
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 192.168.5.1

Pinging 192.168.5.1 with 32 bytes of data:

Reply from 192.168.5.1: bytes=32 time<1ms TTL=255
Reply from 192.168.5.1: bytes=32 time<1ms TTL=255
Reply from 192.168.5.1: bytes=32 time=1ms TTL=255
Reply from 192.168.5.1: bytes=32 time<1ms TTL=255

Ping statistics for 192.168.5.1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 1ms, Average = 0ms

C:\>ping 192.168.3.1

Pinging 192.168.3.1 with 32 bytes of data:

Reply from 192.168.3.1: bytes=32 time<1ms TTL=255
Reply from 192.168.3.1: bytes=32 time<1ms TTL=255
Reply from 192.168.3.1: bytes=32 time<1ms TTL=255
```

PC7

Physical Config Desktop Programming Attributes

Command Prompt

```
Reply from 192.168.3.1: bytes=32 time<1ms TTL=255
Reply from 192.168.3.1: bytes=32 time<1ms TTL=255
Reply from 192.168.3.1: bytes=32 time<1ms TTL=255

Ping statistics for 192.168.3.1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>ping 192.168.2.1

Pinging 192.168.2.1 with 32 bytes of data:

Reply from 192.168.2.1: bytes=32 time<1ms TTL=255
Reply from 192.168.2.1: bytes=32 time<1ms TTL=255
Reply from 192.168.2.1: bytes=32 time=1ms TTL=255
Reply from 192.168.2.1: bytes=32 time<1ms TTL=255

Ping statistics for 192.168.2.1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 1ms, Average = 0ms
```

PC8

Physical Config Desktop Programming Attributes

Command Prompt

```
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 192.168.5.1

Pinging 192.168.5.1 with 32 bytes of data:

Reply from 192.168.5.1: bytes=32 time<1ms TTL=255
Reply from 192.168.5.1: bytes=32 time<1ms TTL=255
Reply from 192.168.5.1: bytes=32 time<1ms TTL=255
Reply from 192.168.5.1: bytes=32 time<1ms TTL=255

Ping statistics for 192.168.5.1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>ping 192.168.3.1

Pinging 192.168.3.1 with 32 bytes of data:

Reply from 192.168.3.1: bytes=32 time<1ms TTL=255
Reply from 192.168.3.1: bytes=32 time=1ms TTL=255
Reply from 192.168.3.1: bytes=32 time=1ms TTL=255
```

PC8

Physical Config Desktop Programming Attributes

Command Prompt

```
Reply from 192.168.3.1: bytes=32 time=1ms TTL=255
Reply from 192.168.3.1: bytes=32 time<1ms TTL=255

Ping statistics for 192.168.3.1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 1ms, Average = 0ms

C:\>ping 192.168.2.1

Pinging 192.168.2.1 with 32 bytes of data:

Reply from 192.168.2.1: bytes=32 time<1ms TTL=255
Reply from 192.168.2.1: bytes=32 time<1ms TTL=255
Reply from 192.168.2.1: bytes=32 time<1ms TTL=255
Reply from 192.168.2.1: bytes=32 time<1ms TTL=255

Ping statistics for 192.168.2.1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>
```